

ANSH SINGH

Prayagraj, India | +91 95557 51574 | ianshsingh10@gmail.com | LinkedIn | GitHub | Portfolio | Codeforces

ABOUT

Full-Stack Developer with hands-on experience building production-grade web applications using the MERN stack, Next.js, and AWS. Delivered AI-integrated platforms in healthcare and GIS domains. Published researcher in IEEE Xplore. Proficient in Java, C++, and Python with applied knowledge of system design, REST API architecture, and CI/CD pipelines. Experienced in technical leadership and large-scale event execution.

EXPERIENCE

AmasQIS.AI

May 2025 – Nov 2025

Full Stack Developer Intern — AI-Integrated GIS & Real Estate Platform

Remote, India

- Developed a full-stack property search and document-verification system serving **2,000+ monthly users**, reducing invalid property listings by **40%** through automated background verification workflows.
- Engineered **RESTful API routes** in Node.js and Express.js that cut backend response time by **30%** and improved throughput for geospatial queries by leveraging spatial indexing in MongoDB.
- Designed **MongoDB schemas** for **10,000+ property records** with optimized geospatial indexes (2dsphere), enabling sub-100ms location-based lookups at scale.
- Integrated a **seller registration pipeline** with backend document verification checks, reducing fraudulent listings and increasing buyer trust metrics on the platform.

The Electric Vehicle Club

Sep 2023 – Mar 2025

Club President

Sehore, India

- Directed club strategy and led a core technical team of **12 members** to execute **5+ technical events**, engaging **1,000+ students** in EV innovation and sustainable technology initiatives.
- Grew student participation in technical workshops by **30%** by coordinating with industry experts and faculty to deliver structured hands-on learning programs.

SKILLS

Languages: Java, C++, Python, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3

Frontend: React.js, Next.js 14, Tailwind CSS, Zustand, Redux, Responsive Web Design

Backend: Node.js, Express.js, REST API Design, Flask, JWT Authentication, OAuth 2.0, NextAuth.js

Databases: MongoDB (Aggregation, Geospatial Indexes), PostgreSQL, MySQL, Redis

Cloud & DevOps: AWS (EC2, S3, IAM), Docker, CI/CD Pipelines (GitHub Actions), Vercel, Nginx, Linux

CS Fundamentals: DSA, System Design, OOP, Operating Systems, DBMS, Computer Network

Tools: Git, GitHub, Postman, VS Code, Figma (basic), Razorpay SDK, Twilio API

Soft Skills: Technical Leadership, Team Management, Event Planning, Active Listening

PROJECTS

craVIT | Next.js 14, MongoDB, Tailwind CSS, Zustand, Razorpay, NextAuth.js | **Live** | **GitHub**

Nov 2025

- Architected a full-stack **Next.js 14** food ordering platform with **Role-Based Access Control (RBAC)** using **NextAuth** and **OAuth 2.0**, rendering distinct dashboards for sellers and buyers from a shared codebase.
- Integrated **Razorpay payment gateway** and **Twilio OTP verification** into the checkout flow, delivering a complete, production-ready e-commerce transaction pipeline.
- Optimized global state management with **Zustand**, synchronizing cart data across the entire UI and reducing unnecessary component re-renders by **40%**, improving frontend performance on low-bandwidth connections.

HealthEase | Node.js, Express.js, React.js, MongoDB, OAuth 2.0, REST API | **IEEE Xplore** | **GitHub**

Apr 2025

- Authored and published** a peer-reviewed research paper detailing the system architecture and measured real-world impact of HealthEase, accepted and indexed in **IEEE Xplore**.
- Built a centralized healthcare web platform consolidating access to a **national hospital database of 200+ facilities**, integrating **5 core healthcare services** — appointment scheduling, blood donation management, ambulance tracking, bed availability, and medical record storage — into a single patient portal.
- Implemented **encryption** for data at rest and in transit, combined with **OAuth 2.0** and **multi-factor authentication (MFA)**, securing sensitive health records for all platform users.
- Developed a custom **AI chatbot** resolving **1,500+ user health queries** and an automated medical report analyzer that handled **70%** of incoming report analysis without manual intervention.
- Integrated **real-time ambulance GPS tracking** and an **SOS emergency alert** system that reduced simulated emergency response time by **45%** in testing scenarios, alongside live hospital bed availability updates.
- Built a real-time **blood donation management module** connecting registered donors with hospitals by blood type, cutting simulated blood shortage incidents by **30%** through inventory forecasting algorithms.
- Achieved **92% user satisfaction rate** in usability studies conducted with healthcare professionals and patients, with users reporting faster access to critical medical information and reduced administrative overhead.

EDUCATION

VIT Bhopal University
B.Tech in Computer Science and Engineering | CGPA: 8.63

Aug 2023 – Ongoing
Sehore, India

CERTIFICATIONS & ACHIEVEMENTS

Certifications: AWS Solutions Architect (Intellipaat, 2025); Introduction to Machine Learning — NPTEL/Swayam (2025); Programming in Java (Vityarthi, 2025); AI/ML (Vityarthi, 2025); Python Essentials (Vityarthi, 2024); Java Programming (Great Learning, 2023); Front-End Development (Great Learning, 2023)

Achievements: 2nd Place — Dev Inferno Hackathon; 1st Place — Face Painting (ALFRESCO Cultural Fest); Published author in **IEEE Xplore** (2025).

Sports: Athletics (2 Gold, 4 Silver, 1 Bronze at school level) Football (Under-16 State Level).